James Bao

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Career Objective

*I am **chasing an internship** to grow my love for everything from **analogue electronics** to **full-stack web development**.*
I want to **improve human lives**—whether through hardware, software, or a **beautiful combination of both**.
I am always looking to **learn** new things and **grow** as a person—and I apply this mindset to everything I do.

Skills Summary

Problem Solving

- I **apply** my strong **logical thinking** to challenging situations, and leave things **better than I found them**.

Communication Skills

- I am an effective communicator with an aptitude for **breaking down challenging concepts**.

Leadership

- I am passionate about intent-based leadership and **understanding** each member's **strengths and weaknesses**.

Teamwork

- I strive to **be an asset** to any team. I am known for my reliability—**I do not make promises I cannot keep**.

Education

Bachelor of Engineering (Honours)

2021 – 2024

The University of Auckland

- **Computer Systems** Engineering
- Cumulative GPA of 8.3/9.0 (**A**–**A+**)
- Dean's Honours List

Experience

Embedded Software Engineer Intern

Nov 2022 – Mar 2023

Windcave Ltd — *Auckland, NZ*

- Writing **C/C++ firmware** on a new STM32U585 **ARM** development board
- **Rapid prototyping** of external **QSPI and OSPI** flash memories
- **Creating** a simple **Point of Sale system** on CHU200TP payment terminal

Freelance Software Developer

2020 – 2021

- Back-end development with JavaScript, **Node.js**, and **MongoDB**
- Built a custom Discord bot tailored to streamline **digital order management** and processing

Technical Skills

Electronics and Embedded Systems

- Analogue **circuit & PCB design** learnt through practical application
- Extensive **hands-on** experience with **soldering** and using **multimeters** & **oscilloscopes**
- Familiarity with **LTspice**, Microchip Studio, Proteus VSM, and **Altium Designer**
- Experience writing **firmware** for **Atmel megaAVR** and **ARM Cortex-M33**

TypeScript/JavaScript

- Front-end **browser** and back-end **Node.js** applications
- Object-oriented and functional patterns
- **async/await**, Promises, and callbacks
- **Serverless functions** with Next.js and Cloudflare Workers

Git/GitHub

- Fluency with **feature branches**, pull requests, issues, and GitHub Actions
- **6100+ contributions** accrued over the past **5 years**

Other Languages I am also experienced with **C/C++**, and have familiarity with **VHDL**, **C#**, **Python**, **Java**, and **MATLAB**.

Project Experience

notion-assignment-import
2022

jamesnzl.xyz/notion-assignment-import

- **Cross-browser** extension that loads Canvas assignments into a Notion database
- **5200+ unique users** in **eighteen months**
- Learnt more about **REST APIs**, HTTP and HTTP clients, and **OAuth2**

- Git and GitHub
- TypeScript, HTML
- Sass and PostCSS
- Gulp, Browserify, and ESBuild
- Insomnia

Smart Energy Monitor
2022

cs209.jamesnzl.xyz — *The University of Auckland*

- **Smart energy monitor** I designed in a **group of four**
- Analogue **circuit & PCB** design, signal conditioning, using an **ATmega328PB**, writing **firmware** drivers, and UART & **Bluetooth** communication

- Lecturer's Choice for **Top Team**
- Git and GitHub
- LTspice
- Microchip Studio
- Proteus VSM
- Altium Designer

Pronto
2020 – 2021

aucklandarmycadets.github.io/Pronto — *City of Auckland Cadet Unit*

- Full-featured Discord bot to **streamline** Unit **procedures**
- Comprehensive **codebase documentation** with **JSDoc**
- Experimenting with **Cloud Computing** services including DigitalOcean, Linode, Vultr, and currently **Oracle Cloud**

- JavaScript, Node.js
- MongoDB
- JSDoc
- Virtual Private Servers
- SSH, Bash, Unix CLI

Unit Website
2022

github.com/aucklandarmycadets/aucklandarmycadets — *City of Auckland Cadet Unit*

- A **modern, performant website** to overhaul my Cadet Unit's online presence
- Created my **own CDN** to cache Notion's ephemeral images and **optimise performance**

- Next.js (and React) with Vercel
- Serverless functions
- DNS Records, SEO

Other Projects More projects can be found on my **GitHub profile**.

Achievements

- **First in Course Award** for ELECTENG/COMPSYS 209: **Analogue and Embedded Firmware Design**
- **First in Course Award** for ENGGEN 131: Introduction to Engineering Computation and **Software Development**
- **Dean's Honours List 2022**
- Bayonet of Honour
- Top of Course Award
  NZCF Senior Non-Commissioned Officer's Course
- School Prefect and Deputy House Captain
- First in Level 3 Calculus
- First in Level 2 Physics
- First in Level 2 Programming

References

Weili Liu
Embedded Development Manager

weili.liu@windcave.com — *Windcave Ltd*

Weili has been a **significant mentor** and **guided** me closely during my internship.

Duleepa J Thrimawithana
Senior Lecturer

d.thrimawithana@auckland.ac.nz — *The University of Auckland*

Duleepa invited me to be a COMPSYS 209 **Teaching Assistant** and to assist with his **Smart Car workshop** at Mānawa Mai **Open Day**, based on my **technical proficiency** and **communication skills**.

Matua Johnny Waititi
Te Kaihautū o Te Puna

jwa@westlake.school.nz — *Westlake Boys' High School*

Matua Johnny was my Te Reo Māori **kaiako** and **form teacher**, and has played a **pivotal** role in my life.